Keep in Mind

This is a practice test. It mirrors the real Unit 5 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

Part A — Vocabulary (8 pts)

Write the letter of the best description next to each term.

1. Index ☐
2. Radicand ☐
3. Principal root ☐
4. Conjugate ☐
5. Extraneous solution ☐
6. Composition ☐
7. Inverse function ☐
8. Anchor point ☐

(A) Everything written underneath the radical symbol; when the index is even, it may not be negative.

(B) The operation that feeds one function's output into another function's input, written $(f \circ g)(x)$.

(C) The small number in the crook of the radical symbol that says which power is being undone.

(D) The point (h, k) that positions a transformed radical graph — an endpoint when the index is even, a turning point when it is odd.

(E) A candidate that satisfies the powered-up equation but fails when it is tested in the original.

(F) The non-negative value the radical symbol by itself returns; a $\pm$ must be written in by hand.

(G) The function that reverses another's inputs and outputs; its graph is the mirror image over the line $y = x$.

(H) A binomial with the same two terms but the opposite middle sign; multiplying by it turns an irrational denominator rational.

Part B — Multiple Choice (12 pts)

Circle the best answer.

1. Written with a rational exponent, $\sqrt[3]{x^5}$ is ☐ (A) $x^{5/3}$ (B) $x^{3/5}$ (C) x^{15} (D) $x^{-3/5}$

2. The domain of $f(x) = \sqrt{7 - 2x}$ is ☐ (A) $x \geq \frac{7}{2}$ (B) $x \geq -\frac{7}{2}$ (C) $x \leq \frac{7}{2}$ (D) all real numbers

3. The graph of $y = \sqrt{3x - 12} + 5$ begins at the point ☐ (A) $(12, 5)$ (B) $(4, -5)$ (C) $(-4, 5)$ (D) $(4, 5)$

4. In simplest form, $\sqrt{75} + \sqrt{12}$ is ☐ (A) $\sqrt{87}$ (B) $7\sqrt{3}$ (C) $7\sqrt{6}$ (D) $10\sqrt{3}$

5. Which equation has *no real solution*? **Letter:** ☐

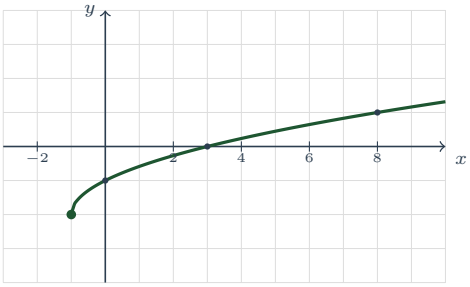
   (A) $\sqrt[3]{x+1} = -4$ (B) $\sqrt{x-1} = 7$ (C) $\sqrt{x+6} = -2$ (D) $\sqrt{x} = 0$

6. If $f(x) = \sqrt{x}$ and $g(x) = x + 9$, then $(f \circ g)(x) =$ ☐ (A) $\sqrt{x} + 9$ (B) $\sqrt{9x}$ (C) $x\sqrt{x} + 9\sqrt{x}$ (D) $\sqrt{x+9}$

Part C — Short Answer & Computation (40 pts)

Show your work. Assume every variable is non-negative unless told otherwise.

1. **Read the graph.** The radical function f is graphed below; each gridline is one unit and the solid dot is the endpoint.



   (a) Coordinates of the endpoint: ☐

   (b) Domain (interval notation): ☐; range: ☐

   (c) x-intercept: ☐; y-intercept: ☐

   (d) Interval(s) where f is increasing: ☐

   (e) Does f have an absolute maximum or an absolute minimum? Give it and say where it occurs.

(f) End behavior: as $x \to \infty$, $f(x) \to$; as $x \to -\infty$,

(g) Interval(s) where $f(x) < 0$:

2. **Radicals and rational exponents.** Rewrite or evaluate. Leave no negative exponents.

(a) Write $\sqrt[4]{x^3}$ with a rational exponent.

(b) Write $y^{5/2}$ as a radical.

(c) $27^{2/3} =$ (d) $16^{-1/2} =$

(e) Write $x^{1/2} \cdot x^{1/3}$ as a single radical. **Answer:**

3. **Simplify. Add or subtract where you can.**

(a) $\sqrt{72}$ (b) $\sqrt[3]{54}$ **(a)** **(b)**

(c) $\sqrt{20x^3}$ (d) $5\sqrt{12} - \sqrt{27}$ **(c)** **(d)**

4. **Multiply, then rationalize.** Every answer must be in simplest form.

(a) $\sqrt{6} \cdot \sqrt{15}$ (b) $\left(2 + \sqrt{5}\right)\left(2 - \sqrt{5}\right)$ **(a)** **(b)**

(c) $\dfrac{6}{\sqrt{3}}$ (d) $\dfrac{4}{3 - \sqrt{5}}$ **(c)** **(d)**

5. **Transformations.** Let $g(x) = -2\sqrt{x+3} + 4$.

(a) List every transformation that takes $y = \sqrt{x}$ to g.

(b) Anchor point: ____

(c) Domain: ____; range: ____

(d) Is the anchor an absolute maximum or an absolute minimum? ____

(e) A different graph of the form $y = \sqrt{x-h} + k$ begins at $(5, -2)$. Write its equation. ____

(f) For $p(x) = \sqrt[3]{x-1} + 2$, give the anchor point ____ and the domain ____. In one sentence, say why that domain is not like g's.

6. **Solve. Test every candidate in the original equation.**

(a) $\sqrt{2x+3} = x$ **Answer:** ____

(b) $\sqrt[3]{x-4} = -2$ **Answer:** ____

(c) $\sqrt{3x-2} + 4 = 1$ **Answer:** ____

(d) $x^{2/3} = 25$ **Answer:** [ ]

7. **Composition.** Let $f(x) = \sqrt{x}$ and $g(x) = x - 6$.

   (a) $(f \circ g)(x) =$ [ ], domain: [ ]

   (b) $(g \circ f)(x) =$ [ ], domain: [ ]

   (c) Are these the same function? Answer using *both* the rule and the domain.

   [ ]

   (d) $(f \circ g)(10) =$ [ ]

   (e) For $p(x) = 2x + 1$ and $q(x) = x^2 - 3$, find $(p \circ q)(-2)$. [ ]

8. **Inverses.**

   (a) $f(x) = 4x - 7$. Find $f^{-1}(x)$. [ ]

   (b) Verify by composition — *both* orders — that your answer really is the inverse.

   [ ]

   (c) $g(x) = \sqrt{x + 2}$. Find $g^{-1}(x)$ and state its domain. $g^{-1}(x) =$ [ ] **domain:** [ ]

   (d) The point $(7, 3)$ lies on the graph of g. What point must lie on the graph of g^{-1}? [ ]

Part D — Extended Response (12 pts)

Explain your reasoning in full sentences.

1. **The full analysis.** Consider $f(x) = \sqrt{x-1} + 2$. **(a)** Give the anchor point, the domain, and the range, both in interval notation. **(b)** Find the x-intercept and the y-intercept, or explain why one does not exist. **(c)** List the transformations that take $y = \sqrt{x}$ to f. **(d)** Find $f^{-1}(x)$ and state its domain, saying exactly where that restriction comes from. **(e)** Justify with composition — both orders — that f and f^{-1} are inverses. **(f)** A classmate writes $y = (x-2)^2 + 1$ with *no* restriction and calls it the inverse. Explain why that is not the inverse of f.

2. **Modeling.** Ignoring friction, a roller-coaster car that starts from rest and drops h feet reaches the bottom at a speed of

$$v(h) = 8\sqrt{h} \text{ feet per second.}$$

(a) State the domain that makes sense here, and connect it to the rule about what an even index will accept. **(b)** Find $v(9)$ and $v(36)$, with units. What does quadrupling the drop do to the speed, and which feature of $\sqrt{\cdot}$ is responsible? **(c)** A car reaches 56 feet per second at the bottom. Find the height of the drop algebraically, and check your answer. **(d)** Write the inverse, h as a function of v, and say what it tells an engineer that $v(h)$ does not. **(e)** Part (c) required squaring both sides. Explain why there is no danger of an extraneous solution in this problem.